

Literature Benchmark Brief: Fedus_2022_JMLR

Title: Switch Transformers: Scaling to trillion parameter models with simple and efficient sparsity

Authors: W. Fedus, B. Zoph, N. Shazeer (2022)

Venue: Journal of Machine Learning Research | **DOI:** 10.5555/3586589.3586709

Cluster: Cluster E: Missing Data & Modular Risk Layers

Core Theoretical Mechanism & Model Formulation

Top-1 routing simplicity, auxiliary balancing loss, distributed capacity factors

Empirical Scale & Experimental Dataset

Trillion-parameter transformer models

Relevance to IEEE TSTE & Sustainable Energy Systems

Exemplar of simplified sparse routing in neural architectures

Identified Gaps & Operational Vulnerabilities

Evaluated purely in natural language; lacks physical boundary constraints

Role in Manuscript Defense & Reviewer #2 Proof

Referenced in Section II-B for routing regularisation design